



PuraBeet®

Version 1.4 replaces V1.3
Revision date: 29.08.2023

Print date: 22.07.2024

Section 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name	PuraBeet®
Chemical name	Betaine
CAS- No.	107-43-7
EC- No.	203-490-6
Registration-No.	01-2119520508-42-0009

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses	Personal care
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1.3 Supplier

	Beta Pura GmbH Josef-Reither-Strasse 21 – 23 A-3430 Tulln T: +43 2272 602 0 Email: office@betapura.com
Competent person	Dr. Paul Meissner Email: office@betapura.com

1.4 Emergency telephone number

+43 2272 602 0
Available during office hours:
Mo – Fr: 8 a.m. – 3 p.m.

Please contact the regional poison center or emergency call.

Austrian toxicity information center Vienna:
+43 1 406 43 43
Available 24 hours

Section 2: Hazards identification

2.1 Classification of the substance or mixture

- **According to regulation (EC) No. 1272/2008**

The substance is not classified as hazardous in sense of the regulation (EC) No. 1272/2008.

2.2 Label elements

- **According to regulation (EC) No. 1272/2008**

Not applicable.

The substance is not classified as hazardous and does not require labelling in sense of the regulation (EC) No. 1272/2008.



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▪ Hazard components for labelling

Not applicable.

2.3 Other hazards

Dust can combine with air to form an explosive mixture.

Results of PBT and vPvB assessment:

PBT: Betaine does not meet the criteria for classification as PBT in sense of REACH.

vPvB: Betaine does not meet the criteria for classification as vPvB in sense of REACH.

Determination of endocrine-disrupting properties

The product does not contain substances with endocrine-disrupting properties ≥ 0.1 % (w/w).

Section 3: Composition/information on ingredients

3.1 Substances

Betaine – Purity > 99% (w/w)

CAS: 107-43-7

EC: 203-490-6

Section 4: First aid measures

4.1 Description of first aid measures

Seek medical advice.

Never give anything by mouth to an unconscious person or a person with cramps.

Change contaminated, saturated clothing.

▪ In case of inhalation of dust

Provide fresh air. Consult a doctor in case of any complaints.

If unconscious, place and transport in recovery position.

▪ In case of skin contact

After contact with skin, wash with plenty of water.

Change contaminated, saturated clothing. Consult a physician if symptoms occur.

▪ In case of eye contact

In case of contact with eyes, flush immediately with plenty of flowing water for 10 to 15 minutes holding eyelids apart. Consult a physician.

▪ In case of ingestion

Rinse mouth with cold water and then drink plenty of water. Avoid vomiting. Consult a physician immediately.

4.2 Most important symptoms and effects, both acute and delayed

No further information available.

4.3 Indication of any immediate medical attention and special treatment needed



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Depending on the condition of the patients, the doctor must assess the symptoms and the overall general condition.

Section 5: Firefighting measures

5.1 Extinguishing media

- **Suitable extinguishing media**

Use firefighting measures that suit the environment.
CO₂, foam, dry chemicals, water spray

- **Extinguishing media which must not be used for safety reasons**

Water jet

5.2 Special hazards arising from the substance or mixture

Fire may cause evolution of CO_x, NO_x, trimethylamine
Dust can combine with air to form an explosive mixture.

5.3 Advice for firefighters

Special protective equipment:

Wear a self-contained breathing apparatus and chemical protective clothing.

Collect contaminated fire extinguishing water separately. Do not allow to enter into drains or surface water.

Section 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Restricted access to contaminated areas, until cleaning work is finished.

Wear personal protection equipment.

Avoid contact with skin and eyes. Provide adequate ventilation.

Avoid formation of dust. Avoid breathing dust.

Keep ignition sources away.

Dust can combine with air to form an explosive mixture.

6.2 Environmental precautions

Do not allow to enter into surface water, groundwater or drains.

6.3 Methods and material for containment and cleaning up

Pick up mechanically.

Avoid formation of dust.

Treat the recovered material as prescribed in the section on waste disposal (section 13)

6.4 Reference to other sections

See protective measures under section 8 and disposal under section 13.

Section 7: Handling and storage



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7.1 Precautions for safe handling

Provide adequate ventilation. Avoid contact with skin and eyes.
Avoid generation of dust – do not breathe dust. Keep containers tightly closed.
Observe protective measures and safety instructions.

7.2 Conditions for safe storage, including any incompatibilities

▪ Precautions against fire and explosion

Keep ignition sources away – Do not smoke.

▪ Requirements on storage and packaging/containers

Keep dry.
Provide adequate ventilation.
Keep containers tightly closed.
Product is hygroscopic. Protect from light.

▪ Incompatibility with materials

No data available.

▪ Conditions of storage

Room temperature

7.3 Specific end use(s)

Personal care.

Section 8: Exposure controls/personal protection

8.1 Control parameters

Customers are advised to check their local regulations.

Valid for Austria & Germany: The product does not contain any relevant quantities of substances with critical values that have to be monitored at the workplace.

DNEL-value (Derived no effect level)

betaine (CAS 107-43-7)

Workers:

No hazards identified.

Consumers:

DNEL long-term, systemic, oral: 4413 mg/kg BW/d

DNEL short-term, systemic, oral: 11178 mg/kg BW/d

PNEC-Werte (Predicted no effect concentration)

betaine (CAS 107-43-7)

Fresh water: 1,2 mg/l

8.2 Exposure control



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▪ General protection and hygiene measures

When handling with chemical substances observe usual precautionary measures.

Keep away from foods and drinks.

When using do not eat, drink, smoke.

Wash hands after working with product.

Avoid contact with skin and eyes.

Avoid formation of dust.

Avoid breathing dust.

Change contaminated clothing before reuse.

Protective clothing needs to be selected specifically for the workplace, depending on concentrations and quantities of the hazardous substances handled. The chemical resistance of the protective equipment should be enquired at the respective supplier.

▪ Respiratory protection

In case of inadequate ventilation wear respiratory protection.

▪ Hand protection

Suitable protective gloves are recommended. EN 374.

Nitrile rubber, butyl rubber, latex, PVC

The selection of suitable gloves depends not only on the material, but also on other quality characteristics. Breakthrough times and swelling properties of the material must be taken into consideration.

▪ Eye protection

Protective goggles. EN 166.

▪ Body protection

Protective clothes.

▪ Environment exposure control measures

Do not allow to enter into surface water, groundwater or drains.

Section 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state / Appearance	solid
Colour	white
Odour	Faint molasses odour
Odour threshold	No information available.
Melting point/freezing point	No information available.
Initial boiling point/range	No information available.
Flammability	No information available.
Lower explosive limits	No information available.



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Upper explosive limits	No information available.
Flash point	n. a.
Ignition temperature	No information available.
Decomposition temperature	No information available.
pH value	n. a.
Viscosity	n. a.
Solubility in water (20 °C)	soluble
Partition coefficient: n-octanol/water	No information available.
Vapour pressure (50 °C)	n. a.
Density	No information available.

9.2 Other information

Form	crystalline
Explosive properties	Not explosive. Fine dust can combine with air to form an explosive mixtures.
Oxidising properties	No information available.
Evaporation rate	No information available.

Information with regard to physical hazard classes

Explosives	Void
Flammable gases	Void
Aerosols	Void
Oxidising gases	Void
Gases under pressure	Void
Flammable liquids	Void
Flammable solids	Void
Self-reactive substances and mixtures	Void
Pyrophoric liquids	Void
Pyrophoric solids	Void
Self-heating substances and mixtures	Void
Substances and mixtures, which emit flammable gases in contact with water	Void



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Oxidising liquids	Void
Oxidising solids	Void
Organic peroxides	Void
Corrosive to metals	Void
Desensitised explosives	Void

Section 10: Stability and reactivity

10.1 Reactivity

No hazardous reactions known if stored and used as prescribed.

10.2 Chemical stability

Product is hygroscopic.

10.3 Possibility of hazardous reactions

Dust explosions hazard.

10.4 Conditions to avoid

Heat, ignition sources
Humidity

10.5 Incompatible materials

Oxidizing agents, alkalis, concentrated

10.6 Hazardous decomposition products

Risk of development of trimethylamine in hot, strongly alkaline solution.

Section 11: Toxicological information

11.1 Information on toxicological effects

▪ LD₅₀ values of single components relevant for classification

Betaine:

LD₅₀ oral (rat): 11179 mg/kg (Source: European Chemicals Agency www.echa.eu)

▪ Acute toxicity

Based on available data classification criteria are not fulfilled.

▪ Skin corrosion/irritation

Based on available data classification criteria are not fulfilled.

▪ Serious eye damage/irritation

Based on available data classification criteria are not fulfilled.

▪ Skin/Respiratory sensitization

Based on available data classification criteria are not fulfilled.



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- **Germ cell mutagenicity**

Based on available data classification criteria are not fulfilled.

- **Cancerogenicity**

Based on available data classification criteria are not fulfilled.

- **Reproductive toxicity**

Based on available data classification criteria are not fulfilled.

- **Specific target organ toxicity – single exposure**

Based on available data classification criteria are not fulfilled.

- **Specific target organ toxicity – repeated exposure**

Based on available data classification criteria are not fulfilled.

- **Aspiration hazard**

Based on available data classification criteria are not fulfilled.

11.2 Information on other hazards

- **Endocrine disrupting properties**

Substance is not listed.

Section 12: Ecological information

12.1 Toxicity

- **Aquatic Toxicity of ingredients**

Betaine:

Daphnia: EC50 (48 h): 4335 mg/l – Daphnia magna

Algae: EC50 (72 h): 1199 mg/l - Desmodesmus subspicatus

(Source: European Chemicals Agency www.echa.eu)

12.2 Persistence and degradability

Readily biodegradable (88% (28 d))

12.3 Bioaccumulative potential

No bioaccumulation potential

12.4 Mobility in soil

No data available.

12.5 Results of PBT and vPvB assessment

Betaine does not meet the criteria for classification as PBT or vPvB in sense of REACH.

12.6 Endocrine disrupting properties

The product does not contain substances with endocrine disrupting properties.

12.7 Other adverse effects



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Do not allow to enter into surface water, groundwater or drains.

Section 13: Disposal considerations

13.1 Waste treatment methods

Any disposal practice must be in compliance with all local and national laws and regulations. Customers are advised to check their local legislation governing the disposal of waste materials.

If this preparation becomes a waste, the final user must define and assign the appropriate European Waste Catalogue code. Use only authorized contractors.
Do not allow to enter into surface water or drains.

- **European waste catalogue** (complies with UVEK valid for CH)

16 03 06 – organic wastes other than those mentioned in 16 03 05

Notes: The European Waste Catalogue (EWC) classifies waste materials and categorises them according to what they are and how they were produced. This may cause other classifications. The final decision belongs to the last user.

- **Uncleaned packaging**

Recommendation: after complete emptying and cleaning, send to an authorized contractor to be reconditioned or recycled.

Section 14: Transport information

No dangerous good in sense of the transport regulation for land, air and sea.

14.1 UN number

Not applicable.

14.2 UN proper shipping name

Not applicable.

14.3 Transport hazard class(es)

Not applicable.

14.4 Packing group

Not applicable.

14.5 Environmental hazards

Not applicable.

14.6 Special precautions for user

Not applicable.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable.



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Section 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Directive 2012/18/EU

Named dangerous substances - ANNEX I

Substance is not listed.

DIRECTIVE 2011/65/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment – Annex II

Substance is not listed.

REGULATION (EU) 2019/1148

Annex I - RESTRICTED EXPLOSIVES PRECURSORS (Upper limit value for the purpose of licensing under Article 5(3))

Substance is not listed.

Annex II - REPORTABLE EXPLOSIVES PRECURSORS

Substance is not listed.

Regulation (EC) No 273/2004 on drug precursors

Substance is not listed.

Regulation (EC) No 111/2005 laying down rules for the monitoring of trade between the Community and third countries in drug precursors

Substance is not listed.

National Regulations:

Germany: WGK 1 (slightly hazardous for water)

15.2 Chemical safety assessment

Not available.

Section 16: Other information

The information contained herein is based on the present state of our knowledge. It characterizes the product with regard to the appropriate safety precautions. It does not represent a guarantee of the properties of the product.

Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist. All materials may present unknown hazards and should be used with caution and only for identified uses as prescribed in section 1.

The classification is based on manufacturer's data and additional information of hazardous substances database and the European Chemicals Agency.

▪ **Department issuing SDS**

UmEnA GmbH
<http://umena.at>

▪ **Date of previous version:** 04.10.2022

▪ **Version number of previous version:** 1.3

▪ **Abbreviations and acronyms**

SAFETY DATA SHEET

according to regulation (EC) 1907/2006 – REACH



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ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)
IMDG: International Maritime Code for Dangerous Goods
IATA: International Air Transport Association
GHS: Globally Harmonised System of Classification and Labelling of Chemicals
EINECS: European Inventory of Existing Commercial Chemical Substances
ELINCS: European List of Notified Chemical Substances
CAS: Chemical Abstracts Service (division of the American Chemical Society)
DNEL: Derived No-Effect Level (REACH)
PNEC: Predicted No-Effect Concentration (REACH)
LD50: Lethal dose, 50 percent
PBT: Persistent, Bioaccumulative and Toxic
vPvB: very Persistent and very Bioaccumulative

▪ **Changes in sections** 2